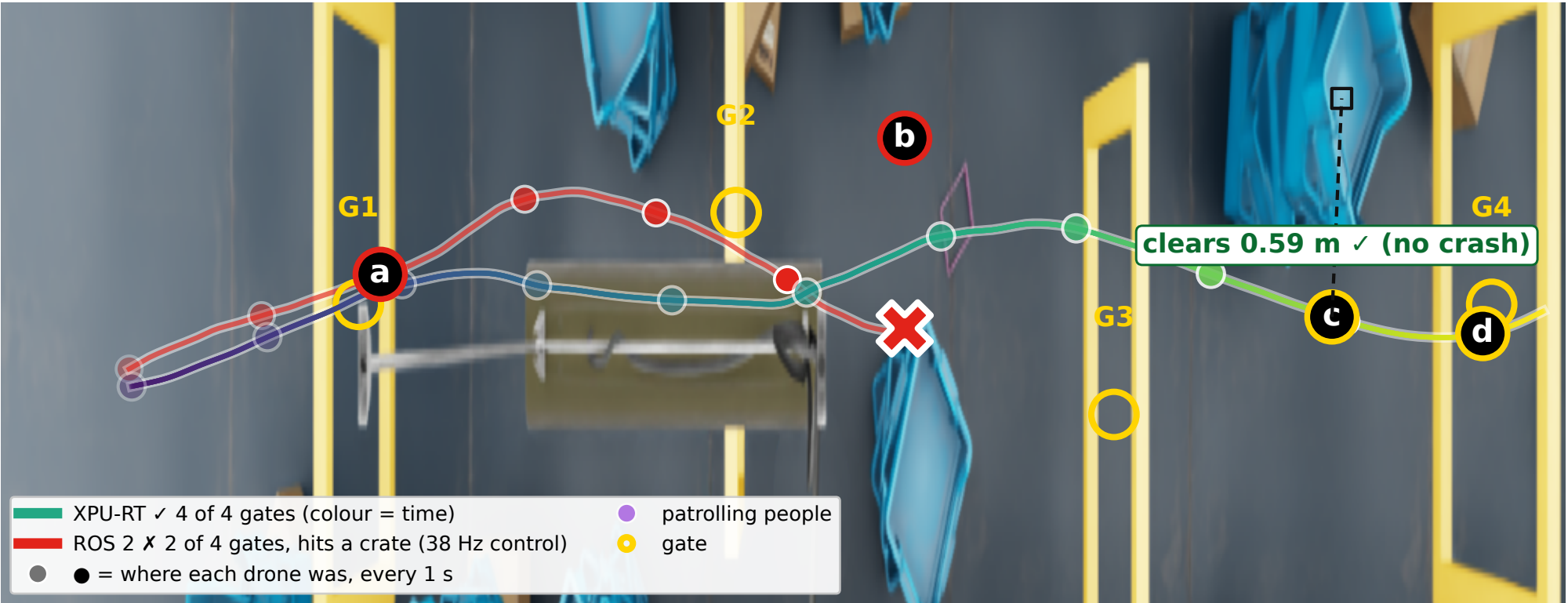
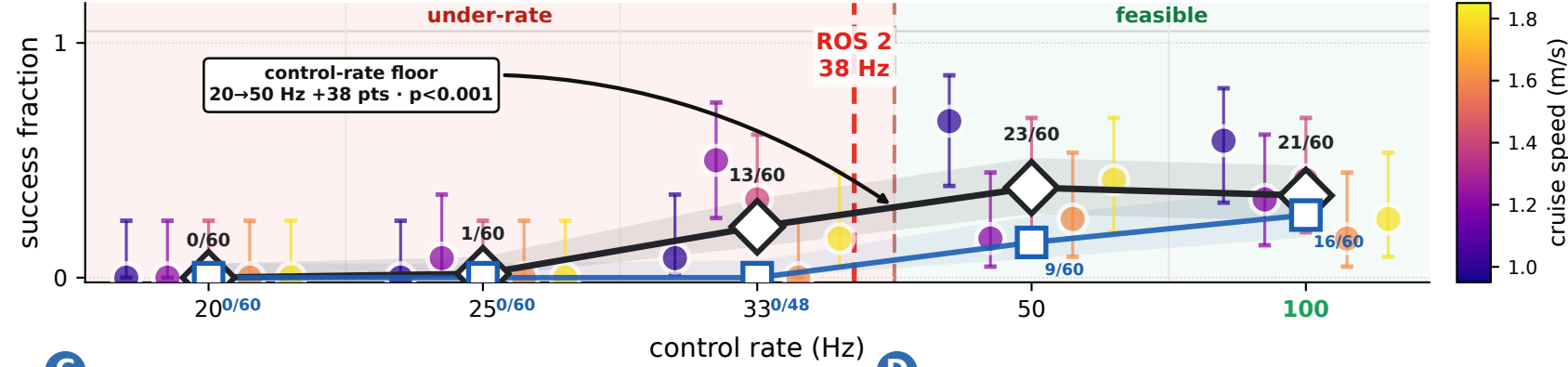


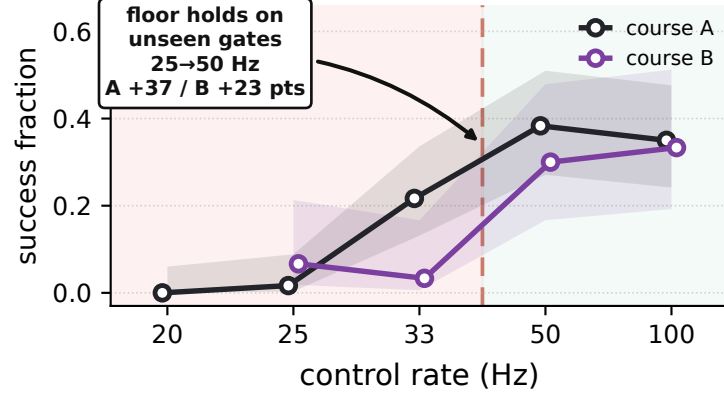
Warehouse gate-course showdown — same scene, drone and gain, at 1.4 m/s
XPU-RT · solver-placed CP-SAT schedule, conv on the IME (103 Hz control) clears all 4 gates
ROS 2 · static 6-core partition, two cores idle (38 Hz control) clears G2 → hits a crate
one seed of this scene (seed 1007); over its 12 seeds XPU-RT completes 3, ROS 2 none



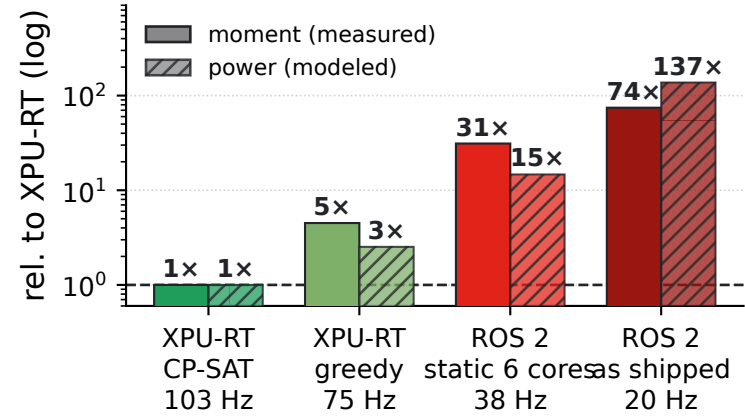
B Control-rate envelope (rate injected, 300 flights): success against control rate — ROS 2's 38 Hz and XPU-RT's 103 Hz marked



C Generalization: unseen gates (course B)



D Mechanism: the baseline thrashes (n=6 flights/arm)



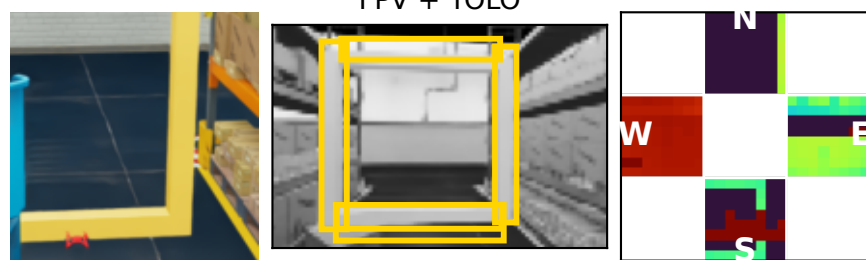
a ROS 2 38 Hz · clears G1 · t=2.8s



b ROS 2 38 Hz · hits crate after G2 · t=6.8s



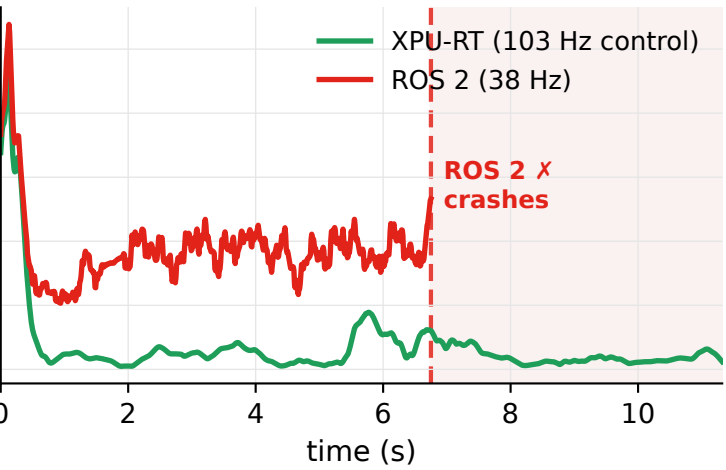
c XPU-RT 103 Hz · clears crate 0.59 m · t=9.8s



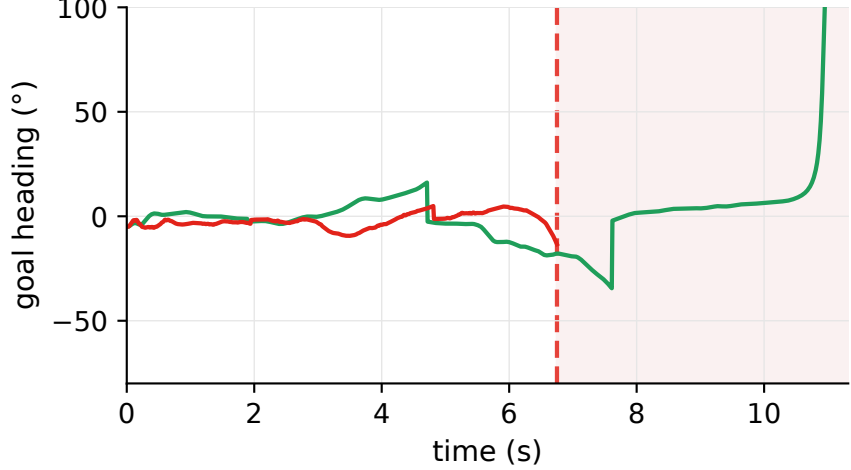
d XPU-RT 103 Hz · reaches gate · t=10.9s



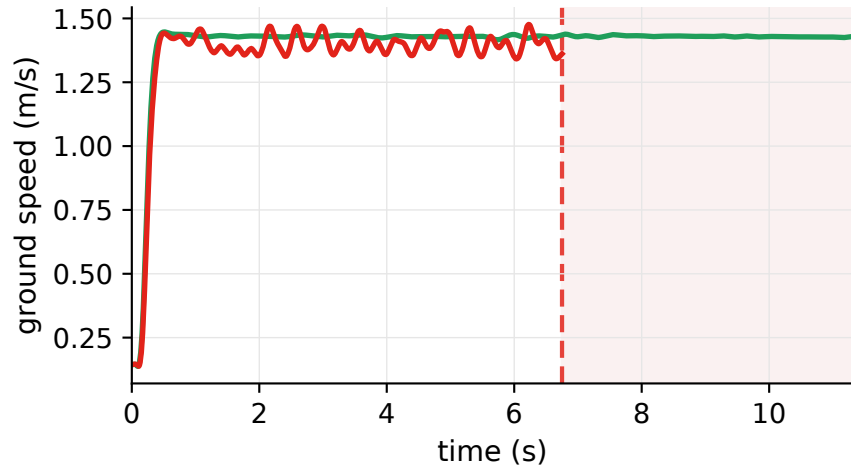
E Inside the pair — body rate: the baseline thrashes at 38 Hz



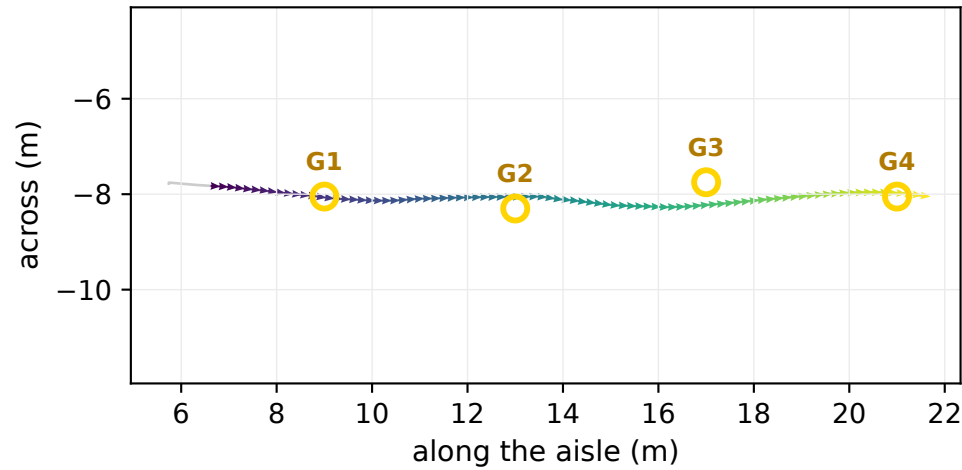
F Nav goal heading: the two commands diverge before the baseline's collision



G Forward speed: the same cruise until the baseline's collision



H XPU-RT's velocity through the gates (arrow = heading · speed, colour = time)



I Onboard K1 schedule, measured at the 45 Hz camera — XPU-RT · solver-placed CP-SAT schedule, conv on the IME places the kernels over all 8 harts: camera→control 27 ms, control every 9.9 ms, every frame on time · YOLO on 8 harts, control on 8 harts, nav on 8 harts
ROS 2 · pinned (3 processes · single executor · chained control · pinned): camera→control 57 ms, control every 25.9 ms, 765/769 frames late · YOLO on 4 harts, 1 traced, control on 1 hart, nav on 1 hart

